

Muhammed Sadique P

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Professional Summary

- 6+ years of experience as an Embedded Hardware Design & Validation Engineer.
 - Proven expertise in designing and validating industrial, commercial, and automotive-grade Microprocessor & Micro controller based boards such as imx6,iMx8,STM32,etc.
 - Proficient in Ethernet, USB, UART, CAN, CAN-FD, RS232/485, I2C, SPI, SERDES, and other interfaces.
 - Skilled in debugging boards, resolving assembly/design issues, and validating hardware and product performance.
 - Experienced in GNSS, Wi-Fi, Bluetooth, Ethernet, and sensor systems (accelerometers, gyroscopes).
 - Expertise in PUMP driver,Solenoid,ADC,etc
 - Extensive knowledge of EMI/EMC and pre-compliance testing.
 - Well-versed in ISO standards (ISO 26262, ISO 16750-1/2/4, CISPR 25) and product development life cycles.
 - Experienced in lab management and utilizing tools like oscilloscopes, logic analysers, and multimeters,
 - Expertise in hardware product validation, reliability testing, and board bring-up.
 - Expertise in product validation, reliability testing and field testing(automotive and industrial) and electromechanical testing.
 - Component Engineering, value engineering, product sustenance, EOL Management.
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Technical Skills

Circuit Design & Verification

- Hardware design and validation (micro controllers,microprocessors,FPGA,etc)
- Analog and mixed-signal design.
- Power circuit estimation, sequencing, and management.

Performance Validation

- Hardware design verification and validation.
- RF and digital interface validation (Wi-Fi, Bluetooth, GNSS, LTE, CAN, CAN-FD).
- Electrical and environmental testing per ISO and IEC standards.
- Field integration testing (vehicle integration, Industrial integration, etc).
- Power Electronics testing.

- System engineering, reliability testing and product validation.

PCB Design

- PCB layout review and support.
- Schematic capture.
- Netlist checking and assembly instructions preparation.
- Circuit designing..

Tools

- ORCAD Schematics Capture, Cadence Allegro Viewer, Tera Term, Hyper Terminal.
 - Bugzilla, and Tortoise SVN and JIRA.
 - JAMA, WINDCHILL, etc
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Professional Experience

Senior Hardware Design & Validation Engineer *June 2019 – June 2024 iWave Systems Technologies Pvt Ltd, Bangalore*

- Designed and validated iMX6 and iMX8-based rugged telematics units and gateways.
- Schematic capture, component selection, and BoM generation.
- Led board bring-up, interface testing, and performance validation.
- Supported PCB layout guidelines, mechanical design integration, and environmental testing.
- Ensured product compliance with CE, FCC, EMI-EMC, and ISO standards.
- Automated testing using hardware-in-loop (HIL) systems.
- Managed reliability testing, product validation, and lab operations.
- Team lead for product validation.

Professional II *July 2024 – present Capgemini Engineering*

- Validated automotive audio amplifier compliance and electromechanical testing.
- Conducted pre-compliance testing and hardware design verification.
- Contributed to design and development architecture of gas station monitoring system.
- Validation of gas station monitor system.
- System integration and testing of multi gas monitoring.
- Experience with monitoring hazardous gas detection and calibration.
- Board bring-up and testing.
- Component engineering and legacy product sustenance engineering.
- Bug tracking and electromechanical testing.

Key Projects

1. iMX6 Rugged Telematic Control Unit (Version 2)

- Designed and validated interfaces including Ethernet, Wi-Fi, Bluetooth, UART, CAN, RS232/485, GNSS, and LTE based iMX6ULL.
- Electromechanical testing and product validation.
- Delivered fully tested boards meeting ISO standards.

2. iMX6 Automotive Telematic Control Unit

- Designed hardware and validated interfaces like accelerometers, gyroscope sensors, and digital I/O.
- Assisted in CE/FCC certification and EMI/EMC testing.

3. iMX8 Automotive Telematic Gateway

- 1G/100M/10M automotive Ethernet and RF interfaces validation.
- Supported certification testing and debugging.
- Team lead for product validation.

4. Low-Cost Telematic Dongle

- Designed systems with Wi-Fi, Bluetooth, CAN, USB, and GNSS interfaces.
- Conducted component selection, and interface testing.

5. Gas station control board based on IMX93 SOM & STM32.

- Designing and validation of gas detection and monitoring controller boards(NFC,BLE,solenoid,pump motor,pressure sensors,wireless charging,USB2,etc)
- System integration and testing.
- Board bring-up and hardware validation.
- Gas flow testing and flow algorithm.
- Electromechanical testing.

Education

- **B.E. in Electronics and Communication Engineering** (*Pursuing*)
 - VVCET Tamil Nadu, Anna University
- **Engineering Diploma**
 - GPTC Manjeri, 2019, 8.99 CGPA
- **PUC**

- GHSS Kavanur, 2015, 85%
- **SSLC**
 - MAOHSS Elayur, 2013, 90%

Achievements

- Received the **“Pat on the Back” award** (2021, 2022) for exceptional contributions to telematic control unit design and validation.

Languages

- English, Malayalam, Kannada, Tamil, Hindi

Permanent Address

Thadathil House, Eliyaparambu, Kavanur PO, Malappuram, Kerala, 673639